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United States Patent	12413062
Kind Code	B2
Date of Patent	September 09, 2025
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Mounting device for securing and routing elongated articles

Abstract

A mounting device configured to attach an elongated article to a substrate includes a base portion and an attachment portion extending from the base portion. The attachment portion is configured to be received within an opening in the substrate and has a plurality of branches extending therefrom. The mounting device also includes a cap portion that is spaced apart from the base portion and defining an aperture in which the attachment portion is received. The plurality of branches engage the aperture and are configured to inhibit removal of the attachment portion from the aperture. The mounting device further includes a plurality of legs extending between the base portion and the cap portion that is configured to retain the elongated article between the base portion and the cap portion. A method of attaching an elongated article to a substrate using such a mounting device is also described.

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Appl. No.: 18/237050

Filed: August 23, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20250070540 A1	Feb. 27, 2025

Publication Classification

Int. Cl.: H02G3/30 (20060101)

U.S. Cl.:

CPC **H02G3/30** (20130101);

Field of Classification Search

CPC: H02G (3/30)

USPC: 174/50

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Background/Summary

TECHNICAL FIELD OF THE INVENTION

(1) This disclosure generally relates to a mounting device for securing elongated articles, such as electrical cables to a substrate and routing the elongated articles.

BACKGROUND

(2) Securing a flat flexible electrical cable (FFC) and/or a flexible printed circuit (FPC) to a substrate such as a body panel of a vehicle, and routing the FFC or FPC, along the substrate an/or through a structure presents problems due to a tendency of the FFC and FPC to buckle if it is not properly contained at the attachment point. Additional issues with FFC and FPC is that they are typically folded over to make turns or bends in a horizontal plane. However, the indexing of the conductor is swapped by the folding. This requires the number and location of folds to be carefully accounted for to guarantee proper indexing of the conductors in the FFC or FPC. Further, the current mounting devices for FFC and FPC are not well suited for automated installation.

SUMMARY

(3) According to one or more aspects of the present disclosure to a mounting device configured to attach an elongated article to a substrate includes a base portion and an attachment portion extending from the base portion that is configured to be received within an opening in the substrate. The attachment portion defines a plurality of branches extending therefrom. The mounting device also includes a cap portion that is spaced apart from the base portion. The cap portion defines an aperture in which the attachment portion is received. The plurality of branches engage the aperture and are configured to inhibit removal of the attachment portion from the aperture. A plurality of

legs extending between the base portion and the cap portion are configured to retain the elongated article between the base portion and the cap portion.

(4) In some aspects of the mounting device described in the previous paragraph, the elongated article is a flat electrical cable.

(5) In some aspects of the mounting device described in any one of the previous paragraphs, the mounting device is configured to retain the flat electrical cable between the base portion, the attachment portion, the cap portion and at least one leg of the plurality of legs.

(6) In some aspects of the mounting device described in any one of the previous paragraphs, the attachment portion and the aperture are sized, shaped and arranged such that the attachment portion and the aperture are aligned in a single particular orientation.

(7) In some aspects of the mounting device described in any one of the previous paragraphs, the attachment portion has a rounded triangular cross-section, and the aperture has a corresponding rounded triangular shape.

(8) In some aspects of the mounting device described in any one of the previous paragraphs, the plurality of legs extend from the cap portion.

(9) In some aspects of the mounting device described in any one of the previous paragraphs, the plurality of legs extend through holes defined in the attachment portion.

(10) In some aspects of the mounting device described in any one of the previous paragraphs, the plurality of legs are jointed and extend between the cap portion and the attachment portion.

(11) In some aspects of the mounting device described in any one of the previous paragraphs, a pair of legs in the plurality of legs have joints that are configured to allow the pair of legs to flex away from each other.

(12) In some aspects of the mounting device described in any one of the previous paragraphs, the plurality of legs extend from the attachment portion.

(13) In some aspects of the mounting device described in any one of the previous paragraphs, the plurality of legs extend from an edge of the attachment portion and beyond the cap portion.

(14) In some aspects of the mounting device described in any one of the previous paragraphs, the substrate serves as the cap portion.

(15) In some aspects of the mounting device described in any one of the previous paragraphs, the mounting device further includes the elongated article which is a first elongated article that is positioned between the base portion, the attachment portion, the cap portion, and m legs of the plurality of legs. The mounting device further includes a second elongated article which is positioned between the base portion, the attachment portion, the cap portion, and $m+n$ legs of the plurality of legs; m and n are nonzero integers.

(16) In some aspects of the mounting device described in any one of the previous paragraphs, the first elongated article has a first radius of curvature around the attachment portion and the second elongated article has a second radius of curvature around the attachment portion that is shorter than the first radius of curvature.

(17) According to one or more aspects of the present disclosure, a method of attaching an elongated article to a substrate includes the steps of: providing a mounting device having a base portion, an attachment portion extending from the base portion. The attachment portion defines a plurality of branches extending therefrom. The mounting device also has a cap portion that is spaced apart from the base portion. The cap portion defines an aperture in which the attachment portion is received. The plurality of branches engage the aperture and are configured to inhibit removal of the attachment portion from the aperture. The attachment portion also has a plurality of legs extending between the base portion and the cap portion that is configured to retain the elongated article between the base portion and the cap portion; disposing the elongated article in the mounting device such that it is positioned between the base portion, the attachment portion, the cap portion, and m legs of the plurality of legs, wherein m is a nonzero integer; and inserting the attachment portion within an opening in the substrate, wherein the plurality of branches engage the opening

and are configured to inhibit removal of the attachment portion from the opening.

(18) In some aspects of the method described in the previous paragraph, the attachment portion and the opening are sized, shaped and arranged such that the attachment portion and the opening are aligned in a single particular orientation.

(19) In some aspects of the method described in any one of the previous paragraphs, the attachment portion has a rounded triangular cross-section, and the opening has a corresponding rounded triangular shape.

(20) In some aspects of the method described in any one of the previous paragraphs, the elongated article is a first elongated article and wherein the method further includes routing a second elongated article through the mounting device such that it is positioned between the base portion, the attachment portion, the cap portion, and $m+n$ legs of the plurality of legs, wherein n is a nonzero integer.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The present invention will now be described, by way of example with reference to the accompanying drawings, in which:

(2) FIG. 1 is an isometric view of mounting device in accordance with an embodiment of the invention;

(3) FIG. 2 is a cross-section view of the mounting device of FIG. 1 in accordance with an embodiment of the invention;

(4) FIG. 3 is an isometric view of the mounting device of FIG. 1 retaining a flat electrical cable in accordance with an embodiment of the invention.

(5) FIG. 4 is an isometric view of the mounting device of FIG. 1 retaining a wider flat electrical cable than shown in FIG. 3 in accordance with an embodiment of the invention;

(6) FIG. 5 is a side isometric view of a mounting device with a plurality of flexible legs extending from a cap portion in accordance with another embodiment of the invention;

(7) FIG. 6 is a cross-section view of the mounting device of FIG. 5 in accordance with another embodiment of the invention;

(8) FIG. 7 is a top isometric view of the mounting device of FIG. 5 in accordance with another embodiment of the invention;

(9) FIG. 8 is an isometric view of mounting device of FIG. 5 with the plurality of flexible legs in a flexed condition in accordance with another embodiment of the invention;

(10) FIG. 9 is an isometric view of the mounting device of FIG. 5 retaining a flat electrical cable in accordance with another embodiment of the invention;

(11) FIG. 10 is an isometric view of a mounting device retaining a flat electrical cable in accordance with yet another embodiment of the invention;

(12) FIG. 11 is an isometric view of the mounting device of FIG. 10 retaining a narrower flat electrical cable than shown in FIG. 10 in accordance with yet another embodiment of the invention;

(13) FIG. 12 is an isometric view of the mounting device of FIG. 10 retaining a plurality of flat electrical cables in accordance with yet another embodiment of the invention;

(14) FIG. 13 is top view of the mounting device of FIG. 12 retaining a plurality of flat electrical cables in which one flat electrical cable has a radius of curvature that is shorter than a radius of curvature of the other flat electrical cable in accordance with yet another embodiment of the invention;

(15) FIG. 14 is top view of the mounting device of FIG. 12 including cable support devices attached to the flat electrical cables to maintain the radii of curvature in accordance with yet another embodiment of the invention;

(16) FIG. 15 is top view of the mounting device of FIG. 14 including a cap portion and substrate in accordance with yet another embodiment of the invention; and
(17) FIG. 16 is an isometric view of the mounting device of FIG. 15 in accordance with yet another embodiment of the invention.

DETAILED DESCRIPTION

(18) FIGS. 1-4 show a first non-limiting example of a mounting device **100** that is configured to secure a flat flexible electrical cable (FFC) and/or flexible printed circuit (FPC) to a substrate **2**, e.g., a body panel of a vehicle, and to help route the FFC and/or FPC, e.g., along the substrate or through a structure. Hereafter, the flat flexible cables and the flexible printed circuits will be generically referred to as cables **4**. The mounting device **100** includes a disc shaped base portion **102** and an attachment portion **104** extending generally perpendicular from the base portion **102**. The attachment portion **104** is configured to be received within an opening **6** in the substrate **2** by a plurality of branches **106** extending from the attachment portion **104**. The plurality of branches **106** curve or are angled away from a free end of the attachment portion **104**. The base portion, **102**, the attachment portion **104** and the plurality of branches **106** form a structure very similar to that commonly referred to as a fir or Christmas tree fastener, so called due to the shape of the barbed protrusions that run down the length of the fastener which resemble the branches of a fir tree. The mounting device **100** also includes a movable cap portion **108** that is spaced apart from the base portion **102**. The cap portion **108** is also disk shaped and defines an aperture **110** in which the attachment portion **104** is received. The cap portion **108** is configured to move along the attachment portion **104**. The plurality of branches **106** engage the edge of the aperture **110** inhibit removal of the attachment portion **104** from the aperture **110** due to the configuration of the plurality of branches **106**. The mounting device **100** further includes a plurality of legs **112** that extend from the cap portion **108**. The plurality of legs **112** is received within holes **114** defined in the base portion **102**. The plurality of legs **112** forms a cage-like structure that is configured to retain the cable **4** between the base portion **102**, the cap portion **108**, and the attachment portion **104**.

(19) The attachment portion **104**, the aperture **110**, and the opening **6** in the substrate **2** are sized, shaped and arranged such that the attachment portion **104**, the aperture **110**, and the opening **6** are aligned in a single particular orientation. In order to provide this feature, the attachment portion **104** has a cross section with a width greater than length. A first width of one end of the cross section is less than a second width at the other end of the cross section. In the illustrated example the attachment portion **104** has a generally triangular cross-section, best seen in FIGS. 1 and 3. The aperture **110** and the opening **6** have a corresponding rounded triangular shape in order to accommodate the plurality of branches **106**. The attachment portion may alternatively have a cross-section shape having a width on one end is less than a width at the other end, such as a trapezoidal shape.

(20) As shown in FIGS. 3 and 4, the distance between the base portion **102** and the cap portion **108** may be adjusted to accommodate cables **4** having different widths. The insulation between conductors in the cable **4** may be slit to gather the conductors, thereby narrowing the width of the cable **4** in the mounting device **100** as shown in FIGS. 3 and 4.

(21) FIGS. 5-9 show a second non-limiting example of a mounting device **200** configured to secure a cable **4** to a substrate **2** and to help route the cable **4**. The mounting device **200** differs from the mounting device **100** mainly in that the plurality of legs **212** extending from the cap portion **208** do not extend through openings in the base portion **202** but rather abuts the base portion **202**. The plurality of legs **212** are flexible and are arranged in pairs that are configured to flex away from one another as the cap portion **208** is moved closer to the base portion **202** as shown by comparing FIGS. 5 and 8.

(22) FIGS. 10-16 show a third non-limiting example of a mounting device **300** configured to secure a cable **4** to a substrate **2** and to help route the cable **4**. The mounting device **300** differs from the

mounting device **100** mainly in that the plurality of legs **312** extend from the edges of the base portion **302** rather than from the cap portion **308**. The plurality of legs **312** is flexible and are outwardly flexed by contact with the edges of the cap portion **308** as shown in FIGS. **10** and **11**.

(23) As shown in FIG. **13-16**, the mounting device **300** may retain more than one cable **4**. The cables **4** may be routed so that a first cable **4A** has a smaller radius of curvature than a second cable **4B**. This may be accomplished by the retaining the second cable **4B** between more legs of the plurality of legs **312** than the first cable **4A**. The first and second cables **4A**, **4B** may also include cable support devices **8** as shown in FIGS. **14-16** to help maintain the radii of curvature of the first and second cables **4A**, **4B**.

(24) The mounting devices **100** and **200** may also be used to retain more than one cable **4**, similarly to the examples shown for mounting device **300** in FIGS. **13-16**.

(25) The mounting devices **100**, **200** shown herein provide multi-directional routing capability for the cables **4** and a unidirectional attachment portion **104**. The access and egress of the cables **4** to and from the mounting devices **100**, **200** at multiple locations allows for a high degree of application flexibility. The single particular orientation of the attachment portion **104** allows the mounting devices **100**, **200** to be oriented in particular directions to facilitate more precise cable routing while minimizing the number of mounting devices **100**, **200** needed. The mounting devices **100**, **200** are configured to allow routing of the cables **4** without folding the cables **4**. The insulation between the wires in the cables **4** may be slit, i.e., singulated, as shown in the various figures to allow the wires in the cables **4** to be bundled together and provide a more compact height of the cables **4** and mounting devices **100**, **200**. Additional features, such as tape tabs or shoulders, may be added to the mounting devices **100**, **200** to restrict lateral movement of the cables **4** if needed.

(26) The examples presented herein are directed to mounting devices **100**, **200** and components thereof configured to cables **4**. However, other embodiments of the mounting devices **100**, **200** may be envisioned that are adapted to retain other elongated items and articles such as fiber optic cables, pneumatic lines, hydraulic lines, cables, ropes, vines, etc., or a combination of any of these.

(27) While this invention has been described in terms of the preferred embodiments thereof, it is not intended to be so limited, but rather only to the extent set forth in the claims that follow. For example, the above-described embodiments (and/or aspects thereof) may be used in combination with each other. In addition, many modifications may be made to configure a particular situation or material to the teachings of the invention without departing from its scope. Dimensions, types of materials, orientations of the various components, and the number and positions of the various components described herein are intended to define parameters of certain embodiments and are by no means limiting and are merely prototypical embodiments.

(28) Many other embodiments and modifications within the spirit and scope of the claims will be apparent to those of skill in the art upon reviewing the above description. The scope of the invention should, therefore, be determined with reference to the following claims, along with the full scope of equivalents to which such claims are entitled.

(29) As used herein, 'one or more' includes a function being performed by one element, a function being performed by more than one element, e.g., in a distributed fashion, several functions being performed by one element, several functions being performed by several elements, or any combination of the above.

(30) It will also be understood that, although the terms first, second, etc. are, in some instances, used herein to describe various elements, these elements should not be limited by these terms. These terms are only used to distinguish one element from another. For example, a first contact could be termed a second contact, and, similarly, a second contact could be termed a first contact, without departing from the scope of the various described embodiments. The first contact and the second contact are both contacts, but they are not the same contact.

(31) The terminology used in the description of the various described embodiments herein is for the purpose of describing particular embodiments only and is not intended to be limiting. As used in

the description of the various described embodiments and the appended claims, the singular forms “a”, “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will also be understood that the term “and/or” as used herein refers to and encompasses all possible combinations of one or more of the associated listed items. It will be further understood that the terms “includes,” “including,” “comprises,” and/or “comprising,” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

(32) As used herein, the term “if” is, optionally, construed to mean “when” or “upon” or “in response to determining” or “in response to detecting,” depending on the context. Similarly, the phrase “if it is determined” or “if [a stated condition or event] is detected” is, optionally, construed to mean “upon determining” or “in response to determining” or “upon detecting [the stated condition or event]” or “in response to detecting [the stated condition or event],” depending on the context.

(33) Additionally, while terms of ordinance or orientation may be used herein these elements should not be limited by these terms. All terms of ordinance or orientation, unless stated otherwise, are used for purposes distinguishing one element from another, and do not denote any particular order, order of operations, direction or orientation unless stated otherwise.

Claims

1. A mounting device configured to attach an elongated article to a substrate, said mounting device comprising: a base portion; an attachment portion extending from the base portion, configured to be received within an opening in the substrate, and having a plurality of branches extending therefrom; a cap portion spaced apart from the base portion and defining an aperture in which the attachment portion is received, wherein the plurality of branches engage the aperture and are configured to inhibit removal of the attachment portion from the aperture; and a plurality of legs extending between the base portion and the cap portion configured to retain the elongated article between the base portion and the cap portion.
2. The mounting device according to claim 1, wherein the elongated article is a flat flexible electrical cable.
3. The mounting device according to claim 1, wherein the elongated article is a flexible printed circuit.
4. The mounting device according to claim 2, wherein the mounting device is configured to retain the flat electrical cable between the base portion, the attachment portion, the cap portion and at least one leg of the plurality of legs.
5. The mounting device according to claim 1, wherein the attachment portion and the aperture are sized, shaped and arranged such that the attachment portion and the aperture are aligned in a single particular orientation.
6. The mounting device according to claim 5, wherein the attachment portion has a rounded triangular cross-section, and the aperture has a corresponding rounded triangular shape.
7. The mounting device according to claim 1, wherein the plurality of legs extend from the cap portion.
8. The mounting device according to claim 7, wherein the plurality of legs extend through holes defined in the attachment portion.
9. The mounting device according to claim 7, wherein the plurality of legs are jointed and extend between the cap portion and the attachment portion.
10. The mounting device according to claim 9, wherein a pair of legs in the plurality of legs have joints that are configured to allow the pair of legs to flex away from each other.
11. The mounting device according to claim 1, wherein the plurality of legs extend from the

attachment portion.

12. The mounting device according to claim 11, wherein the plurality of legs extend from an edge of the attachment portion and beyond the cap portion.

13. The mounting device according to claim 11, wherein the substrate serves as the cap portion.

14. The mounting device according to claim 1, further comprising: the elongated article which is a first elongated article that is positioned between the base portion, the attachment portion, the cap portion, and m legs of the plurality of legs; and a second elongated article which is positioned between the base portion, the attachment portion, the cap portion, and $m+n$ legs of the plurality of legs, wherein m and n are nonzero integers.

15. The mounting device according to claim 14, wherein the first elongated article has a first radius of curvature around the attachment portion and the second elongated article has a second radius of curvature around the attachment portion that is shorter than the first radius of curvature.

16. A method of attaching an elongated article to a substrate, comprising: providing a mounting device having: a base portion, an attachment portion extending from the base portion and having a plurality of branches extending therefrom, a cap portion spaced apart from the base portion and defining an aperture in which the attachment portion is received, wherein the plurality of branches engage the aperture and are configured to inhibit removal of the attachment portion from the aperture, and a plurality of legs extending between the base portion and the cap portion configured to retain the elongated article between the base portion and the cap portion; disposing the elongated article in the mounting device such that it is positioned between the base portion, the attachment portion, the cap portion, and m legs of the plurality of legs, wherein m is a nonzero integer; and inserting the attachment portion within an opening in the substrate, wherein the plurality of branches engage the opening and are configured to inhibit removal of the attachment portion from the opening.

17. The method according to claim 16, wherein the attachment portion and the opening are sized, shaped and arranged such that the attachment portion and the opening are aligned in a single particular orientation.

18. The method according to claim 17, wherein the attachment portion has a rounded triangular cross-section, and the opening has a corresponding rounded triangular shape.

19. The method according to claim 16, wherein the elongated article is a first elongated article and wherein the method further includes routing a second elongated article through the mounting device such that it is positioned between the base portion, the attachment portion, the cap portion, and $m+n$ legs of the plurality of legs, wherein n is a nonzero integer.

20. The method according to claim 16, wherein the elongated article is selected from a list consisting of a flat flexible electrical cable and a flexible printed circuit.
